

Notes: Adamopoulos and Restuccia (AER, 2014)

The Size Distribution of Farms and International Productivity Differences

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Cross-country farm-size data	4
3.2	Farm size and productivity by size	4
3.3	Aggregate factors	5
3.4	Farm-size distortions	5
4	Model	5
4.1	Two-sector structure	5
4.2	Farm technology and farmer heterogeneity	6
4.3	Preferences and subsistence demand	6
4.4	Equilibrium and average farm size	6
4.5	Interpretation of distortions	6
5	Calibration and quantitative experiments	7
5.1	Benchmark calibration	7
5.2	Aggregate factors experiment	7
5.3	Generic farm-size distortions experiment	7
5.4	Empirical farm-size distortions	8
5.5	Specific policy experiments	8
6	Identification and interpretation	8
6.1	What the paper identifies	8
6.2	Why farm size is central	9
6.3	Role of the U.S. benchmark	9
6.4	Empirical distortion assumptions	9

6.5	Potential amplification channels	9
7	Tables and figures	9
7.1	Figure 1 and Table 1: Farm size and development	9
7.2	Figures 2 and 3: The farm-size distribution	10
7.3	Tables 2 and 3: U.S. farm-size facts and calibration	11
7.4	Figures 4 and 5: Model fit by farm size	11
7.5	Tables 4 and 5: Aggregate factors	12
7.6	Table 6 and Figure 6: Farm-size distortions	13
7.7	Tables 7 and 8: Empirical distortions and land reform	14
8	Short summary	15
9	Conclusion	16
9.1	Core conclusion	16
9.2	Economic intuition	16
9.3	Incremental contribution revisited	16
9.4	Implications	16
9.5	Limitations	16
9.6	Takeaway	17

1 Big picture

- **Research question.** Why are farms much smaller in poor countries, and how much do farm-size differences contribute to international differences in agricultural and aggregate productivity?
- **Main empirical fact.** Rich countries operate agriculture at much larger scale. The poorest countries have average farm size of about 1.6 hectares, while the richest countries have average farm size of about 54.1 hectares, a 34-fold gap.
- **Why farm size matters.** Larger farms have much higher labor productivity. In the U.S. farm data used for calibration, value added per worker is about 16 times higher in the largest farm-size category than in the smallest category.
- **Core mechanism.** The paper builds a two-sector quantitative model in which heterogeneous farmers choose capital and land. Distortions that favor small farms misallocate land and capital away from high-ability farmers, lowering average farm size and measured agricultural productivity.
- **Main quantitative result.** Measured aggregate factors—land, capital, and economy-wide productivity—account for only about one-quarter of the rich-poor differences in farm size and agricultural productivity. Farm-size distortions can account for much of the remaining gap.
- **Empirical distortion evidence.** Using crop-level policy distortions from the World Bank Distortions to Agricultural Incentives database, the authors construct a cross-country measure of farm-size distortions. This measure, combined with aggregate factors, accounts for about half of the rich-poor gaps in farm size and productivity.
- **Policy message.** Some land and agricultural policies in poor countries are effectively “pro-small.” Land ceilings, progressive land taxes, input subsidies targeted at smallholders, and related frictions can reduce farm size and productivity by preventing productive farmers from operating at efficient scale.

2 Incremental contribution of the paper

- **From agricultural productivity gaps to farm-size distributions.** The paper shifts attention from the level of agricultural productivity alone to the organization of production across farms.
- **A distribution of farms instead of a representative farm.** Agriculture is modeled as a sector with heterogeneous farm operators. This lets the authors study how resource allocation across farms affects measured productivity.
- **Quantitative discipline from farm microdata.** The benchmark economy is calibrated to match the U.S. farm-size distribution and input patterns by farm size, including the decline in capital-land ratios with scale.

- **Cross-country accounting.** The model separates what can be explained by aggregate factors from what requires farm-level distortions.
- **Observable distortion measure.** Rather than relying only on hypothetical misallocation wedges, the paper constructs an empirical measure from crop-specific price distortions and crop-specific farm sizes.
- **Policy case studies.** The paper also studies concrete land policies: the Philippine land ceiling under the Comprehensive Agrarian Reform Program and progressive land taxation in Pakistan.

3 Data and measurement

3.1 Cross-country farm-size data

The main cross-country farm-size evidence comes from the 1990 World Census of Agriculture, which reports numbers of agricultural holdings and land area by size category. The authors combine these data with aggregate productivity measures from the Penn World Table and agricultural output, employment, and land measures from Prasada Rao.

- **Countries.** The main cross-country dataset contains 63 countries in 1990.
- **Income groups.** Countries are ranked by real GDP per capita and grouped into income quintiles.
- **Farm definition.** The paper treats an agricultural holding as a farm: an economic unit of agricultural production under single management.
- **Core comparison.** The richest and poorest quintiles differ by about 21-fold in real GDP per capita, 19-fold in real GDP per worker, and 47-fold in real agricultural GDP per worker.

3.2 Farm size and productivity by size

Because the World Census of Agriculture does not report output and inputs by farm size, the paper uses U.S. Census of Agriculture data to discipline farm-level patterns.

- **U.S. farm data.** The calibration relies heavily on the 2007 U.S. Census of Agriculture.
- **Labor productivity.** Value added per worker rises sharply with farm size.
- **Land productivity.** Value added per acre falls with farm size, consistent with the familiar inverse relationship between farm size and land yields.
- **Capital intensity.** The capital-land ratio falls with farm size, even though larger farms use more total capital per farm.

3.3 Aggregate factors

The paper studies three measured aggregate factors:

- **Land:** arable land per worker.
- **Capital:** the capital-output ratio.
- **Economy-wide productivity:** total factor productivity common to the agricultural and nonagricultural sectors.

These factors naturally affect farm size because a country with higher aggregate productivity needs fewer workers in agriculture to satisfy food demand, thereby increasing land per farm.

3.4 Farm-size distortions

The paper studies distortions that reduce the operating scale of high-productivity farmers relative to low-productivity farmers.

- **Generic distortion.** The model first introduces a farm-size tax schedule that penalizes larger, more productive farms.
- **Empirical distortion measure.** The paper then uses crop-level nominal rates of assistance from the World Bank Distortions to Agricultural Incentives database.
- **Key construction.** The authors correlate crop-specific taxes in poor countries with typical U.S. crop-specific farm size. Crops produced at larger scale in the United States are taxed more heavily in poor countries relative to the United States.
- **Policy examples.** The paper studies land ceilings in the Philippines and progressive land taxes in Pakistan.

4 Model

4.1 Two-sector structure

The economy has agriculture and nonagriculture. A representative household consumes both goods and allocates workers across sectors. Agriculture is special because it is produced by heterogeneous farmers, not by a representative farm.

Nonagricultural output is produced using a standard constant-returns technology:

$$Y_n = AK_n^\alpha N_n^{1-\alpha},$$

where A is economy-wide productivity.

4.2 Farm technology and farmer heterogeneity

A farmer with managerial ability s operates a farm using capital k and land ℓ . Agricultural output is produced with decreasing returns to scale:

$$y_a(s) = A\kappa [\theta k^\rho + (1 - \theta)(s\ell)^\rho]^{\gamma/\rho}.$$

- A is economy-wide productivity.
- κ is agriculture-specific productivity.
- s is idiosyncratic farmer ability.
- γ captures decreasing returns or span of control.
- ρ governs the elasticity of substitution between capital and land.

High-ability farmers optimally operate larger farms. Thus, a distribution of abilities generates a distribution of farm sizes.

4.3 Preferences and subsistence demand

The representative household has preferences over agricultural and nonagricultural consumption:

$$u(c_a, c_n) = \phi \log(c_a - \bar{a}) + (1 - \phi) \log(c_n),$$

where $\bar{a}$ captures a subsistence requirement for food. This is important because poor countries must allocate many workers to agriculture when productivity is low.

4.4 Equilibrium and average farm size

In equilibrium, labor, capital, land, and goods markets clear. The household allocates workers across sectors until the wage in nonagriculture equals average profits in agriculture. Average farm size is mechanically linked to the land endowment and the agricultural employment share:

$$AFS = \frac{L}{N_a}.$$

A low level of aggregate productivity or a high food labor requirement raises N_a and therefore lowers average farm size.

4.5 Interpretation of distortions

A farm-size distortion is a policy or friction that prevents more productive farmers from expanding. In the model, this operates like an idiosyncratic tax on larger or higher-ability farmers. The result is a reallocation of land and capital away from high-productivity farms toward low-productivity farms.

The economic force is the same as in the misallocation literature: when factors are not allocated to their highest-productivity uses, aggregate measured productivity falls.

5 Calibration and quantitative experiments

5.1 Benchmark calibration

The benchmark economy is calibrated to the United States. The distribution of farmer ability is assumed log-normal and chosen to match the U.S. farm-size distribution. Other parameters are chosen to match U.S. aggregate and farm-level targets, including the employment share in agriculture, capital-output ratio, average farm size, agricultural income shares, and the capital-land ratio across farm sizes.

Interpretation. The calibrated model is not simply fitting aggregate agriculture. It is disciplined by the whole size distribution and by how inputs and productivity vary across farm sizes.

5.2 Aggregate factors experiment

The first quantitative experiment feeds poor-country values of aggregate factors into the benchmark model.

- Land differences alone barely change the main outcomes.
- Adding lower economy-wide productivity and capital-output ratios raises the agricultural employment share and lowers farm size.
- However, aggregate factors explain only about one-quarter of the observed rich-poor gaps in farm size and agricultural labor productivity.

Interpretation. Poor countries have smaller farms partly because they have lower aggregate productivity and therefore more labor in agriculture. But this is not enough to explain the enormous gaps.

5.3 Generic farm-size distortions experiment

The paper then adds a farm-size distortion calibrated to match the 34-fold rich-poor average farm-size gap. This distortion reallocates factors from large, productive farmers to small, less productive farmers.

The model with aggregate factors plus farm-size distortions can match the employment share in agriculture, average farm size, and agricultural productivity gap between rich and poor countries.

Interpretation. If policies and institutions in poor countries disproportionately constrain large farms, they can generate large productivity losses.

5.4 Empirical farm-size distortions

The paper next asks whether observable policy distortions have enough quantitative power to matter.

- The authors use crop-specific nominal rates of assistance from the World Bank DAI data.
- They use U.S. crop-specific average farm sizes to rank crops by typical operational scale.
- They find a positive correlation: crops that are produced at larger scale in the United States tend to be more heavily taxed in poor countries relative to the United States.
- Feeding this empirical distortion schedule into the model accounts for about half of the rich-poor differences in farm size and productivity.

Interpretation. The empirical measure does not capture every land-market friction or farm-size policy, so accounting for half of the gap is a quantitatively large effect.

5.5 Specific policy experiments

The paper examines two concrete policies.

- **Philippine land reform.** The 1988 Comprehensive Agrarian Reform Program imposed a 5-hectare ceiling on land ownership. In the model, the ceiling reduces average farm size and agricultural labor productivity by about 7 percent before aggregate changes are added.
- **Pakistan progressive land tax.** The 1976 reform exempted farms under 5 hectares and raised taxes on larger farms. The model predicts smaller farms and lower agricultural productivity.

Interpretation. Policy effects can be masked in time-series data if aggregate productivity or capital accumulation rises at the same time. Observing productivity growth after a reform does not mean the reform itself was productivity-enhancing.

6 Identification and interpretation

6.1 What the paper identifies

The paper does not estimate a single causal treatment effect in the way a micro reduced-form paper would. Instead, it uses a disciplined quantitative model to ask how much observed farm-size differences and policy distortions can account for international productivity gaps.

6.2 Why farm size is central

Farm size matters because it summarizes the operational scale at which farmers use land and capital. If productive farmers are prevented from expanding, farm size remains low and factors are misallocated.

6.3 Role of the U.S. benchmark

The U.S. benchmark is used to discipline the distribution of farmer ability and the relation between farm size, capital-land ratios, and productivity. This is powerful but also requires caution: farm production in poor countries may differ in crops, technology, geography, and institutions.

6.4 Empirical distortion assumptions

The empirical farm-size distortion measure assumes that crop-specific farm-size rankings in the United States are informative about typical scale elsewhere. It also captures only policy distortions that show up in measured crop-level prices. It does not fully capture land-market frictions, insecure property rights, credit constraints, or informal constraints on farm expansion.

6.5 Potential amplification channels

The model abstracts from some mechanisms that could amplify the effects of distortions, such as selection of workers into agriculture, human-capital accumulation, endogenous technology adoption, and dynamic productivity growth. If these channels matter, the reported effects of farm-size distortions could be conservative.

7 Tables and figures

7.1 Figure 1 and Table 1: Farm size and development

Read:

- Figure 1 plots average farm size against real GDP per capita across countries.
- The fitted relationship is positive: richer countries tend to have larger farms.
- Table 1 shows that average farm size rises from 1.6 hectares in the poorest quintile to 54.1 hectares in the richest quintile.
- The table also shows that the rich-poor scale gap appears within major crop and livestock categories, not only in total farm size.

Economic message: The size distribution of farms is a core development fact. Rich-country agriculture is not merely more productive; it is organized at a much larger operational scale.

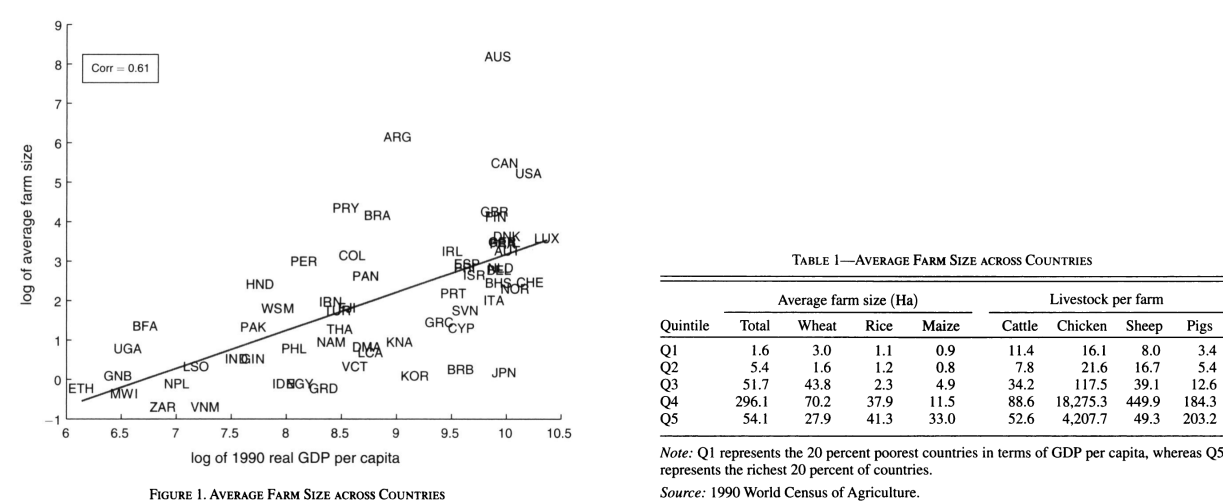


Figure 1: Average farm size across countries.

Table 1: Average farm size across countries.

7.2 Figures 2 and 3: The farm-size distribution

Read:

- Figure 2 shows that poor countries have many small farms and almost no large farms.
- In the poorest quintile, farms below 5 hectares dominate. In the richest quintile, farms above 20 hectares account for a much larger share.
- Figure 3 shows that the land distribution also shifts with development.
- In poor countries, small farms hold a large share of land; in rich countries, large farms hold most of the land.

Economic message: The difference is not just a few outlier farms. The whole distribution of production units changes with development.

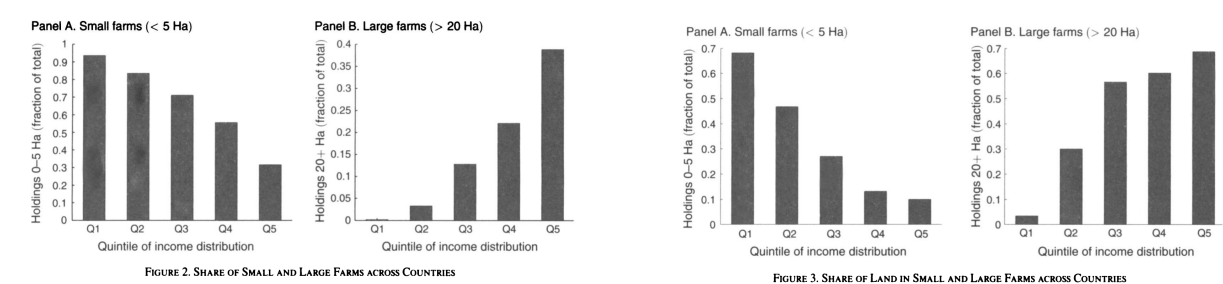


Figure 2: Share of small and large farms across countries.

Figure 3: Share of land in small and large farms.

7.3 Tables 2 and 3: U.S. farm-size facts and calibration

Read Table 2:

- Table 2 reports U.S. farm-size statistics by acres.
- Value added per worker rises strongly with farm size.
- Value added per acre and the capital-land ratio fall with farm size.

Read Table 3:

- Table 3 reports calibrated parameters and targets.
- The ability distribution parameters are chosen to match the U.S. farm-size distribution.
- The agricultural technology parameters are disciplined by factor shares and the capital-land ratio by farm size.

Economic message: The calibration is designed to reproduce not only average U.S. agriculture but also the cross-sectional organization of farms by size.

TABLE 2—STATISTICS BY FARM SIZE IN THE UNITED STATES

Farm size (acres)	Farm distribution	Land share	Value added per acre	Value added per worker	Capital-land ratio
1–9	0.1056	0.0012	33.31	1.00	84.85
10–49	0.2813	0.0173	6.54	1.10	17.88
50–69	0.0698	0.0097	4.23	1.54	9.65
70–99	0.0871	0.0171	3.2	1.92	7.49
100–139	0.0794	0.022	2.67	2.22	5.96
140–179	0.0633	0.0238	2.4	2.67	4.98
180–219	0.0397	0.0187	2.59	3.38	4.73
220–259	0.031	0.0176	2.76	4.15	4.56
260–499	0.0964	0.0823	2.7	5.63	4.05
500–999	0.0679	0.1129	2.92	10.03	3.54
1,000–1,999	0.042	0.1384	2.52	14.25	2.95
2,000+	0.0365	0.5389	1.00	16.45	1.00

Notes: Value added per acre and capital-to-land ratio are normalized relative to the maximum range values. Value added per worker is normalized relative to the minimum range value.

Source: Authors' calculations with data from the 2007 US Census of Agriculture.

TABLE 3—PARAMETERIZATION

Parameter		Value	Target
Technology	A	1	Normalization
	κ	1	Normalization
	α	0.33	Nonagricultural capital income share
	γ	0.54	Agricultural capital income share
	θ	0.89	Agricultural land income share
	ρ	0.24	Capital-land ratio between min-max sizes
Preferences	$\bar{a}$	0.035	Current employment share in agriculture
	ϕ	0.010	Long-run employment share in agriculture
Ability distribution	μ	−1.83	Size distribution
	σ	4.66	Size distribution
Endowments	K	3.9	Aggregate capital-output ratio
	L	4.2	Average farm size
Barrier	ξ	0.66	Value productivity ratio

Table 2: Statistics by farm size in the United States.

Table 3: Parameterization.

7.4 Figures 4 and 5: Model fit by farm size

Read:

- Figure 4 compares the calibrated model to U.S. data for the farm-size distribution and capital-land ratios.
- The model closely matches the broad shape of the size distribution and the decline in capital-land ratios with size.
- Figure 5 compares model and data for land shares, value added per worker, and value added per hectare.
- The model captures the key qualitative pattern: labor productivity rises with farm size while land productivity falls.

Economic message: The model has enough cross-sectional structure to study misallocation across farms, rather than treating agriculture as a representative farm.

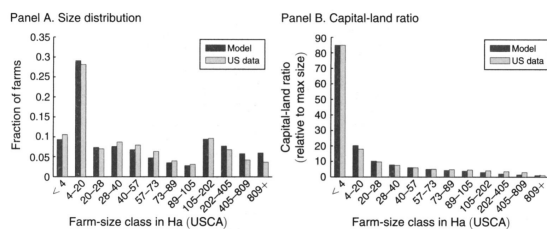


FIGURE 4. CALIBRATED VARIABLES BY FARM SIZE

Figure 4: Calibrated variables by farm size.

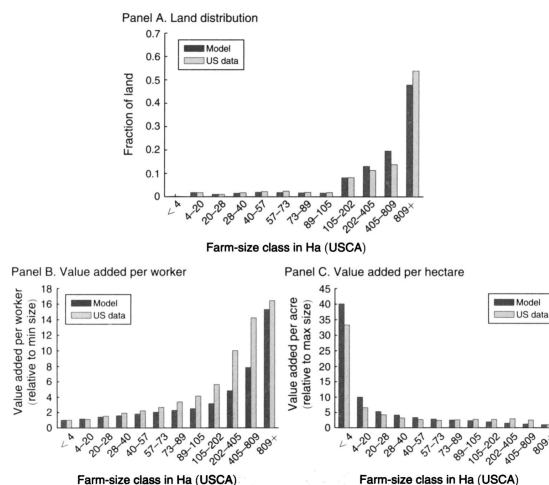


Figure 5: Other variables by farm size.

7.5 Tables 4 and 5: Aggregate factors

Read Table 4:

- Table 4 asks how much land, TFP, and capital differences explain rich-poor gaps.
- Land differences alone have small effects.
- Adding TFP and capital differences raises the agricultural employment share and creates an 8.6-fold farm-size gap, but the data show a 34-fold gap.
- Aggregate factors generate an 11.2-fold agricultural productivity gap, compared with 46.7-fold in the data.

Read Table 5:

- Table 5 applies the same aggregate-factor logic to rich countries.
- Aggregate factors explain much of the variation among rich countries.
- The same factors are much less successful for rich-poor comparisons.

Economic message: Aggregate development matters, but it is not sufficient. Something within the agricultural sector must explain the remaining rich-poor differences.

	Benchmark economy	+ Land differences	+ (TFP, capital) differences	Data
Employment in agriculture N_a (percent)	2.5	2.6	16.6	65.3
Average farm size (AFS)	—	1.3	8.6	34
Labor productivity in agriculture (Y_a/N_a)	—	1	11.2	46.7
Aggregate labor productivity (Y/N)	—	1	7.6	19.2

Notes: Average farm size, real labor productivity in agriculture, and aggregate labor productivity are reported as the ratio between the benchmark economy and the poor economy. Land differences represent the difference in arable land per capita between rich and poor countries which is 1.3-fold. TFP and capital-to-output differences represent a factor of 2.5 and 2.9 between rich and poor countries.

Table 4: Effects of aggregate factors.

	N_a		AFS		Y_a/N_a		Y/N	
	Model	Data	Model	Data	Model	Data	Model	Data
Belgium	2.92	1.86	19.2	16.1	1.31	1.16	1.06	1.05
Netherlands	3.28	3.32	11.1	17	1.56	1.16	1.19	1.18
Germany	3.17	3.46	21.2	32.8	1.49	2.73	1.25	1.27
France	2.71	5.46	80.5	31.5	1.15	1.90	1.07	1.10
Japan	3.31	6.26	5.9	1.2	1.59	14.8	1.23	1.29
Canada	2.74	3.12	336	242	1.17	1.41	1.30	1.31
Denmark	3.17	4.6	77.7	37.8	1.49	1.49	1.36	1.37
Austria	2.81	5.93	42.9	26.4	1.23	2.89	1.08	1.11
Norway	2.79	5.26	41.1	10	1.21	4.74	1.09	1.13
Switzerland	2.74	3.85	12.3	11.8	1.17	2.73	1.02	1.04
Luxembourg	2.26	1.74	22.3	36.2	0.82	1.16	0.73	0.73
Rich (Q5)	2.77	3.93	61.9	42.1	1.19	1.88	1.09	1.11
Poor (Q1)	16.6	65.3	8.6	34	11.2	46.7	7.6	19.2

Notes: N_a is the share employment in agriculture (percent), AFS is average farm size, and Y_a/N_a and Y/N are real labor productivity in agriculture and aggregate labor productivity reported as the ratio between the benchmark economy and each country.

Table 5: Effects of aggregate factors in rich countries.

7.6 Table 6 and Figure 6: Farm-size distortions

Read Table 6:

- Table 6 adds a farm-size distortion to the aggregate-factor model.
- The model matches the 34-fold average farm-size gap by construction.
- It also closely matches the agricultural employment share and agricultural productivity gap.

Read Figure 6:

- Figure 6 plots crop-specific tax rates in poor countries relative to the United States against U.S. average farm size by crop.
- The upward slope means that products produced at larger operational scale in the United States tend to face higher relative taxes in poor countries.
- This provides empirical discipline for a farm-size distortion schedule.

Economic message: A distortion that systematically penalizes large-scale agriculture can produce large measured productivity losses.

	Aggregate factors	+Farm-size distortions	Data
Employment in agriculture N_a (percent)	16.6	65.2	65.3
Average farm size (AFS)	8.6	34	34
Labor productivity in agriculture $(\frac{Y_a}{N_a})$	11.2	46.5	46.7
Aggregate labor productivity $(\frac{Y}{N})$	7.6	17.0	19.2

Notes: Average farm size, real labor productivity in agriculture, and aggregate labor productivity are reported as the ratio between the benchmark economy and the poor economy. Farm-size distortions represent the model with aggregate factors and farm-size distortions given by the tax function $P(s) = 1 - 1/\exp(\psi s)$, where $\psi = 406$ is calibrated to match the factor difference in average farm size of rich relative to poor countries.

Table 6: Effects of farm-size distortions.

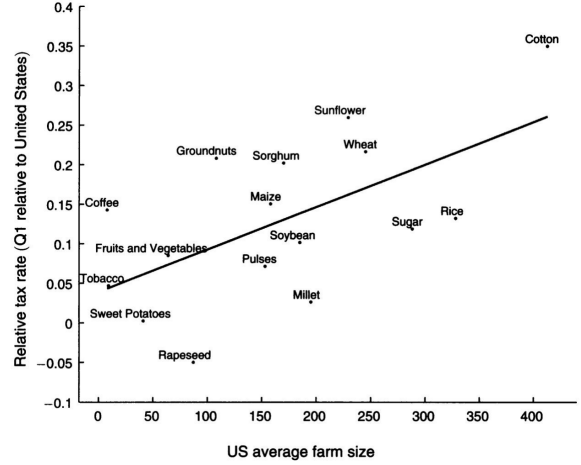


FIGURE 6. EMPIRICAL FARM-SIZE DISTORTIONS

Figure 6: Empirical farm-size distortions.

7.7 Tables 7 and 8: Empirical distortions and land reform

Read Table 7:

- Table 7 feeds the empirical farm-size distortion measure into the model.
- The empirical distortions raise the farm-size gap from 8.6 to 17.3 and the agricultural productivity gap from 11.2 to 23.5.
- A flat tax with the same average tax rate has little effect, showing that the covariance between taxes and farm size is central.

Read Table 8:

- Table 8 studies the Philippine land reform.
- A land ceiling alone reduces average farm size and agricultural labor productivity.
- Changes in aggregate land, capital, and TFP can mask or offset the policy effect in time-series data.

Economic message: The paper's central mechanism is not high average taxation of agriculture by itself. What matters is whether policy distortions are size-dependent and push resources away from productive large farms.

TABLE 7—EFFECTS OF EMPIRICAL MEASURE OF FARM-SIZE DISTORTIONS

	Farm-size distortions	Flat tax	Data
Employment in agriculture N_a (percent)	33.2	15.9	65.3
Average farm size (AFS)	17.3	8.3	34
Labor productivity in agriculture $\left(\frac{Y_a}{N_a}\right)$	23.5	11.1	46.7
Aggregate labor productivity $\left(\frac{Y}{N}\right)$	9.4	7.6	19.2

Notes: Average farm size, real labor productivity in agriculture, and aggregate labor productivity are reported as the ratio between the benchmark economy and the poor economy. Farm-size distortions represent the model with aggregate factors and empirical farm-size distortions from NRA data (with $\psi_0 = 0.73$ and $\psi_1 = 4.9$). Flat tax represents an economy with aggregate factors and a common tax to all farms equal to the mean in the empirical farm-size distortions.

Table 7: Effects of empirical farm-size distortions.

TABLE 8—PHILIPPINE LAND REFORM: CHANGES 1988–2000

	Ceiling	+ ΔL	+ $\Delta\left(\frac{K}{Y}, A\right)$	Data
Employment in agriculture N_a (percent)	48.5	48.6	43.0	37.6
Percentage change in:				
Average farm size (AFS)	−7.0	−32.1	−23.2	−29.6
Labor productivity in agriculture $\left(\frac{Y_a}{N_a}\right)$	−7.0	−7.2	5.2	9.5
Aggregate labor productivity $\left(\frac{Y}{N}\right)$	−5.8	−6.0	13.4	16.4

Notes: ΔL denotes the change in land per capita; $\Delta\left(\frac{K}{Y}, A\right)$ denotes changes in capital-to-output ratio and economy-wide productivity.

Table 8: Philippine land reform.

8 Short summary

- Farm sizes differ enormously across countries: farms are much smaller in poor countries.
- The richest-poorest farm-size gap is about 34-fold, while the agricultural labor productivity gap is about 47-fold.
- U.S. data show that labor productivity rises sharply with farm size.
- The paper builds a two-sector model with heterogeneous farmers and endogenous farm sizes.
- Aggregate factors—land, capital, and economy-wide productivity—explain only about one-quarter of the rich-poor farm-size and productivity gaps.
- Generic farm-size distortions can reproduce the observed rich-poor differences by reallocating land and capital away from productive farmers.
- Observable crop-level distortions explain about half of the farm-size and productivity gaps when combined with aggregate factors.
- The most important distortion is not the average tax level; it is the positive relation between the distortion and operational scale.
- Land ceilings and progressive land taxes can reduce farm size and agricultural productivity.
- Time-series evidence can understate these policy effects if aggregate productivity and capital accumulation move in the opposite direction.

9 Conclusion

9.1 Core conclusion

Adamopoulos and Restuccia show that farm-size differences are central to understanding agricultural productivity gaps. Poor countries have many small farms not only because they have lower aggregate productivity, but also because policies and institutions can prevent productive farmers from operating at larger scale.

9.2 Economic intuition

In an efficient allocation, high-ability farmers operate larger farms and use more resources. Pro-small policies, land ceilings, progressive taxes on large holdings, and related frictions push land and capital toward less productive farms. This reduces average farm size and measured labor productivity.

9.3 Incremental contribution revisited

The paper connects development accounting, agricultural productivity, and the misallocation literature through the size distribution of farms. Its key innovation is to quantify how much of the productivity gap can be explained by farm-level distortions, using both a model and observable policy data.

9.4 Implications

Policies that protect or subsidize small farms may have distributional or political motivations, but they can carry large productivity costs. Improving agricultural productivity in poor countries may require better-functioning land markets, secure property rights, and fewer constraints on the expansion of productive farms.

9.5 Limitations

The empirical farm-size distortion measure captures only price-based distortions and relies on U.S. crop-size rankings. The model abstracts from selection into agriculture, dynamic investment, human capital, and many institutional details. These omissions are important for interpretation but do not overturn the main quantitative lesson that size-dependent distortions can be large.

9.6 Takeaway

The enduring takeaway is that farm size is not a side fact of development. It is a window into resource allocation. When institutions keep farms small, they can substantially depress agricultural productivity and, through structural transformation, aggregate productivity.